

CAPTURE PLAN		2	3	4	5	6
A	USB-C + CC/VBUS protection STUSB4500QTR PD sink and NVM service BQ25798RQMR 1S NVDC charger / power path Battery TEMP safety ownership LTC2955-controlled TPS61288LRQQR 5 V boost TPS62162DSGR 3.3 V logic rail Bulk capacitance, rail discharge and power-good BLOCKED VALUES: battery NTC network, final TVS, magnetics and compensation review.					A
B						B
C						C
D						D
1	2	3	4	5	6	

NOT FOR FABRICATION

Sheet: /01 USB-C, PD, charger and rails/
File: 01_usb_pd_power.kicad_sch

Title: USB-C, PD, CHARGER AND RAILS

Size: A4

Date: 2026-09-05

Rev: 1.0-PHASE1

KiCad E.D.A. 10.0.6

Id: 2/9

NOT FOR FABRICATION		
Sheet: /01 USB-C, PD, charger and rails/ File: 01_usb_pd_power.kicad_sch		
Title: USB-C, PD, CHARGER AND RAILS		
Size: A4	Date: 2026-09-05	Rev: 1.0-PHASE1
KiCad E.D.A. 10.0.6		Id: 2/9

CAPTURE PLAN

1100 Motex 5054402651, 26 positions
Pins 2/4: P4_5V; pins 5/6/16: GND
Four UART pairs from measured ECN-0002 map
Pin 19: SYNC_MASTER; pin 10: CAM_PWR_EN
Pin 21: BTN_SHUTTER_N per ECN-0003
Pins 23/25: shared I2C
Pins 20/22/24/26: C6 service

GPIO35 / PIN 15: DO NOT CONNECT.
3.3 V pins are sense/service only; never back-power the Guition board.

NOT FOR FABRICATION

Sheet: /02 Guition 26-pin interface/
File: 02_p4_interface.kicad_sch

Title: GUITION 26-PIN INTERFACE

Size: A4	Date: 2026-09-05	Rev: 1.0-PHASE1
KiCad E.D.A. 10.0.6	Id: 3/9	

1	2	3	4	5	6	
A	CAPTURE PLAN PAC1954T–E/4MX and four Kelvin shunts Four TPS2553DRVR current–limited branches TCA9539PWR enables and FAULT_N inputs SN74LVC08APWR gates every branch with CAM_PWR_EN Per–branch effective bulk >= 22 uF plus local 100 nF CAM1_5V through CAM4_5V rail test points INITIAL LIMIT: about 1 A; final resistor from measured inrush/capture current.					A
B						B
C						C
D	NOT FOR FABRICATION Sheet: /03 Camera power and monitoring/ File: 03_camera_power.kicad_sch Title: CAMERA POWER AND MONITORING Size: A4 Date: 2026–09–05 Rev: 1.0–PHASE1 KiCad E.D.A. 10.0.6 Id: 4/9					D
1	2	3	4	5	6	

1	2	3	4	5	6	
A	CAPTURE PLAN Eight 3.3 V UART signals with 33 ohm configuration footprints and test pads CAM1 52/51; CAM2 50/49; CAM3 34/33; CAM4 30/29 SN74LVC125APWRG3 four-way SYNC fan-out from GPIO32 / JP1 pin 19 33 ohm branch source resistors, isolation links and test pads CAMn_SYNC to XIAO D1 / GPIO2 CAMn_FSIN, CAMn_PWDN and CAMn_RESET future-interposer pads Route every branch over continuous L2 ground and validate concurrent 921600 baud.					A
B						B
C						C
D	NOT FOR FABRICATION Sheet: /04 UART and synchronization/ File: 04_uart_sync.kicad_sch Title: UART AND SYNCHRONIZATION Size: A4Date: 2026-09-05 KiCad E.D.A. 10.0.6 Rev: 1.0-PHASE1 Id: 5/9					D
1	2	3	4	5	6	

Bench requirement: graceful software shutdown plus 8–10 s forced off with P4 frozen.

Sheet: /05 Power button and shutdown/
File: 05_power_control.kicad_sch

Size: A4	Date: 2026-09-05	Rev: 1.0-PHASE1
KiCad E.D.A. 10.0.6		Id: 6/9

	1	2	3	4	5	6
A	CAPTURE PLAN					
	BQ27441DRZR-G1A plus 10 milliohm Kelvin shunt					
	INA228AIDGSR plus 5 milliohm SYS_5V Kelvin shunt					
	24AA025E48T-I/OT board EEPROM and EUI-48					
	Shared MB_I2C_SDA/SCL configuration links and DNP pull-ups					
	Address straps, write protect, alert routing and test pads					
	Battery NTC is owned by charger safety loop; gauge uses internal or host temperature until curve is known.					
B						
C						
D						
	1	2	3	4	5	6

NOT FOR FABRICATION

Sheet: /06 Fuel gauge, I2C and board ID/
File: 06_monitoring_id.kicad_sch

Title: FUEL GAUGE, I2C AND BOARD ID

Size: A4	Date: 2026-09-05	Rev: 1.0-PHASE1
KiCad E.D.A. 10.0.6		Id: 7/9

1	2	3	4	5	6	
A	CAPTURE PLAN Bottom-side pogo pads for all required rails, UARTs, sync branches and I2C Camera enable/fault and system power-management test access XIAO UART boot/reset recovery routes for CAM1–CAM4 C6 UART, IO9 and CHIP_PU service pads STUSB NVM and board EEPROM production access Serial number and QR/UID programming record Pogo geometry is blocked by mounting coordinates and fixture access.					A
B						B
C						C
D	NOT FOR FABRICATION Sheet: /07 Test, programming and service/ File: 07_test_service.kicad_sch Title: TEST, PROGRAMMING AND SERVICE Size: A4Date: 2026–09–05Rev: 1.0–PHASE1 KiCad E.D.A. 10.0.6Id: 8/9					D
1	2	3	4	5	6	

1	2	3	4	5	6	
A	CAPTURE PLAN USB CC ESD, VBUS TVS/filter and controlled input FETs Battery fuse, reverse polarity and pack-PCM coordination Molex 26-pin Guition and 3-pin battery connectors JST SH shutter connector and ESD/filter option Four identical locking camera connectors Connector pin-1, polarity and harness drawing references BLOCKED SELECTIONS: final cell protection, USB TVS, camera connector/interposer and USB-C receptacle.					A
B						B
C						C
D	NOT FOR FABRICATION Sheet: /08 Protection and connectors/ File: 08_protection_connectors.kicad_sch Title: PROTECTION AND CONNECTORS Size: A4 Date: 2026-09-05 Rev: 1.0-PHASE1 KiCad E.D.A. 10.0.6 Id: 9/9					D
1	2	3	4	5	6	